

SUPPORTLYTICS

IT Support Analytics & Performance Intelligence

A comprehensive analytical evaluation of enterprise IT service desk performance, multi-channel support ticket lifecycles, semantic issue clustering, SLA compliance adherence, and regional operational efficiency. Engineered with Python, advanced DAX measures, and deployed through an interactive 6-page Microsoft Power BI dashboard.

Project Attribute	Enterprise Specification & Deliverable Details
Project Classification	Internship Project — IT Support Analytics
Primary Deliverable	Microsoft Power BI Interactive Dashboard (Supportlytics_Dashboard.pbix)
Analytical Scope	10,000 Support Tickets 38 Schema Attributes 3 Core Clusters 3 Support Tiers
Key Verified Baseline KPIs	Avg Turnaround: 61.1032 hrs SLA Met: 50.39% Similarity Index: 0.7446
Technology Stack	Microsoft Power BI Desktop, DAX, Python 3.10+, Pandas, NumPy, Jupyter
Authoritative Data Source	Supportlytics_PowerBI_Data.xlsx (Sheet: Data)
Documentation Version & Date	Version 1.0 Final Enterprise Release August 2026

PAGE 02 | EXECUTIVE SUMMARY

Strategic Overview of Supportlytics Intelligence System

1. Strategic Project Context & Operational Mandate

Supportlytics is an enterprise analytics and business intelligence solution developed as an Internship Project in IT Support Analytics. Modern enterprise IT support departments handle thousands of service requests, incident reports, and systemic technical problems monthly across distributed global regions. Without centralized intelligence, IT leadership faces severe blind spots regarding operational bottlenecks, SLA breach liabilities, queue imbalances, and repetitive issue overhead. This project establishes an automated data validation and analytics pipeline culminating in a production-ready Microsoft Power BI dashboard (Supportlytics_Dashboard.pbix).

2. Verified Core Key Performance Indicators (KPIs)

KPI Metric	Verified Value	Calculation & Business Definition	Operational Target Status
Total Ticket Volume	10,000 Tickets	COUNTROWS('Data') across multi-channel intakes	Baseline Demand Assessed
Avg Resolution Time	61.1032 Hours	AVERAGE('Data'[Resolution_Hours]) creation to close	Operational Baseline Benchmark
Cluster Similarity Index	0.7446 (74.46%)	AVERAGE('Data'[Similarity_Score]) across clusters	High Semantic Cohesion (> 0.70)
SLA Compliance Rate	50.39% (5,039)	DIVIDE(SLA Met [5,039], Total Tickets [10,000])	5,039 Met / 4,961 Breached
Avg Customer Satisfaction	2.9818 / 5.00	AVERAGE('Data'[Customer_Satisfaction]) post-close	Baseline Service Quality Level
High/Critical Unresolved	2,504 Tickets	CALCULATE(Unresolved, Priority IN {High, Critical})	Active Pending Backlog Focus

3. Key Operational Findings & Executive Takeaways

- 1. Turnaround Metric: Average resolution time stands at 61.1032 hours across all tiers, providing a consistent operational benchmark across the 10,000-ticket log.
- 2. Contractual SLA Distribution: With a 50.39% SLA compliance rate (5,039 Met tickets), 4,961 tickets exceeded contracted resolution windows, highlighting opportunities for queue optimization.
- 3. Semantic Clustering & Deflection Opportunity: Three verified issue clusters (Login, Network, Software) exhibit an average similarity score of 0.7446, supporting knowledge base expansion.

[EXECUTIVE FOCUS] Centralized dashboard monitoring and prioritized management of the 2,504 High/Critical unresolved tickets provide direct operational visibility for service desk leads.

PAGE 03 | PROJECT OVERVIEW

Domain Architecture, Scope Boundaries & Target Users

1. Background and Domain Context

In the contemporary enterprise, the IT Service Desk functions as the primary operational backbone, ensuring continuous availability, hardware reliability, network integrity, access security, and software support for global personnel. As organizations scale, multi-channel intakes (Email, Web Portal, Live Chat, Phone) create voluminous ticket streams. Without rigorous business intelligence, support teams operate reactively—treating each incident in isolation rather than leveraging macro-level behavioral patterns to optimize team allocation, identify underlying systemic defects, and prevent recurring breaches.

2. Analytical Focus & Methodology

Supportlytics bridges operational IT support workflows with rigorous data engineering and visual business intelligence. The project encompasses data cleaning, structured feature engineering, exploratory statistical profiling, semantic similarity analysis, multi-tier team performance benchmarking, spatial geographic mapping, and executive dashboard modeling.

3. Target Stakeholders & Operational Value

Stakeholder Role	Primary Operational Responsibility	Supportlytics Decision Support Value
IT Operations Director	Overall ITSM governance, resource allocation, SLA adherence	Macro KPI visibility, regional benchmarking, breach liability tracking
Service Desk Manager	Daily queue balancing, escalation management, team productivity	Workload distribution, priority monitoring, high-urgency backlog triage
Tier 1/2 Team Leads	Incident resolution speed, first-contact resolution, agent guidance	Turnaround benchmarks across L1, L2, Security teams, category friction points
Continuous Improvement Lead	Problem management, recurring issue reduction, deflection	Cluster analysis (Login, Network, Software), recurring pattern identification

4. Primary Project Deliverables

- Cleaned & Enriched Dataset: Standardized 10,000-ticket dataset with 38 attributes in Supportlytics_PowerBI_Data.xlsx.
- Power BI Production Dashboard: 6-page interactive report (Supportlytics_Dashboard.pbix) with dynamic cross-filtering.
- DAX KPI Architecture: Robust measure library calculating real-time volume, velocity, similarity, SLA, and CSAT metrics.
- Jupyter Analytics Modules: Executable Python notebooks establishing exploratory, clustering, and geographic baselines.

PAGE 04 | BUSINESS PROBLEM

Operational Bottlenecks & Service Delivery Vulnerabilities

1. The Enterprise IT Support Challenge

IT Service Desks manage heavy ticket traffic to minimize downtime, expedite incident resolution, and uphold contracted Service Level Agreements (SLAs). However, operational fragmentation often obscures the root causes of service delays. Analysis of the enterprise support dataset highlights five core operational challenges:

2. Five Core Vulnerability Areas

Vulnerability Area	Observed Empirical Evidence	Operational Risk & Business Consequence
Extended Resolution Duration	Average turnaround duration of 61.1032 hours across 10,000 tickets	Prolonged user waiting time, delayed operational workflows
SLA Non-Compliance Ratio	50.39% SLA met rate (4,961 out of 10,000 tickets breached)	Potential contractual penalties, lower customer satisfaction ratings (2.9818)
High-Priority Backlog Volume	2,504 High and Critical unresolved tickets in active queue	Elevated operational risk from pending high-urgency system issues
Repetitive Incident Overhead	Similarity score averaging 0.7446 across Login, Network, Software clusters	Manual engineering effort spent handling recurring access and software tickets
Geographic Workload Differences	Variations across North America, EMEA, and APAC regions	Resource allocation differences across geographic support queues

3. Why Static Reporting Is Insufficient

Traditional service desks often rely on periodic static summaries or spreadsheet trackers. These mechanisms provide retrospective views rather than interactive diagnostic capabilities. Static reports cannot dynamically cross-filter resolution times across categories, isolate country-specific patterns, or track unresolved high-priority queues in real time.

[CENTRALIZED ANALYTICS IMPERATIVE] An interactive business intelligence dashboard enables support leadership to evaluate workload distribution and queue health dynamically across categories, teams, and geographic locations.

PAGE 05 | PROJECT OBJECTIVES

Quantitative Goals & Technical Deliverables

1. Primary Project Objective

The overarching goal of the Supportlytics project is to engineer an end-to-end data analytics and business intelligence system that converts 10,000 raw IT support ticket records into actionable operational insights, empowering service desk leaders to systematically monitor resolution duration, evaluate SLA compliance, identify recurring issue patterns, and optimize queue management.

2. Specific Quantitative & Technical Objectives

Objective Domain	Technical Scope & Methodology	Expected Deliverable & Success Criteria
Data Standardisation	Ingest and cleanse 10,000 records across 38 operational schema attributes	Zero missing primary keys (Ticket_ID), normalized ISO timestamps, verified types
Feature Engineering	Derive quantitative duration (Resolution_Hours), Priority_Score, Date keys	Engineered analytical features ready for multi-dimensional time intelligence
Exploratory Profiling	Profile distributions across Request (3,358), Problem (3,342), Incident (3,300)	Quantified operational demand across 5 categories and 4 intake channels
Semantic Clustering	Evaluate similarity scores across Login, Network, Software clusters	Cluster Similarity Index (0.7446) established for recurring issue analysis
Performance Benchmarks	Measure turnaround across L1, L2, Security teams, priorities, and countries	Baseline 61.1032-hour turnaround benchmarked across support dimensions
Geographic Analysis	Map ticket density across North America, EMEA, APAC, and 5 countries	Spatial concentration and regional performance visibility established
Power BI Deployment	Build 6-page interactive dashboard with DAX KPI engine and dynamic slicers	Production-grade Power BI dashboard (Supportlytics_Dashboard.pbix) verified

3. Operational Alignment

Each technical objective directly supports key service desk operational priorities: workload monitoring, queue transparency, turnaround tracking, and SLA visibility. The resulting artifacts provide a permanent, reusable decision-support system.

PAGE 06 | SCOPE & ANALYTICAL WORKFLOW

End-to-End Analytics Pipeline Architecture

1. Project Scope Boundaries

- In-Scope: Processing the authoritative 10,000-ticket dataset, 38 schema columns, exploratory data analysis, feature engineering, cluster and similarity analysis, DAX measure modeling, Star-Schema relationship design, Power BI report authoring, and analytical recommendations.
- Out-of-Scope: Direct live API streaming from external ticketing systems and production deployment of automated bot agents.

2. End-to-End Analytical Workflow Pipeline

Pipeline Stage	Core Technical Activities & Transformation Logic
1. Raw Data Ingestion	Acquisition of authoritative enterprise support log in Excel/CSV format (10,000 records).
2. Data Validation & Cleansing	Schema verification, missing value check, type casting, Ticket_ID uniqueness validation.
3. Feature Engineering	Derivation of Resolution_Hours, Priority_Score, Created_Year, Created_Month, Created_Month_Number.
4. Exploratory Data Analysis	Statistical profiling of ticket types, categories, intake channels, departments, and priority ratios.
5. Cluster & Similarity Analysis	Similarity score evaluation across Login, Network, and Software issue clusters.
6. Performance & SLA Modeling	Turnaround duration computation across support teams (L1, L2, Security), countries, and SLA status.
7. Power BI Model & DAX Engine	Creation of star schema (Data fact table + Calendar dimension) and DAX measure calculation engine.
8. Interactive Dashboard & Insights	Authoring of 6 specialized dashboard pages, KPI validation, and strategic insight formulation.

[PIPELINE CONSISTENCY] The analytical workflow ensures consistency: metrics in the final Power BI dashboard reconcile directly with the underlying dataset and DAX calculations.

PAGE 07 | DATASET OVERVIEW

Authoritative Source Structure & Field Group Taxonomy

1. Authoritative Data Source Specification

The definitive data source for this project is Supportlytics_PowerBI_Data.xlsx (Sheet: Data), containing exactly 10,000 records and 38 attributes. Every record represents a distinct IT support ticket logged within the enterprise ecosystem. A 100% unique primary key check confirms exactly 10,000 distinct Ticket_ID values with zero nulls or duplicates.

2. Eight Logical Schema Field Groups

Field Group	Included Dataset Attributes	Operational & Analytical Role
1. Ticket Lifecycle	Ticket_ID, Created_Date, Resolution_Date, First_Response_Date, Status, Resolution_Code, Resolution_Notes	Tracks ticket progression, turnaround duration, lifecycle milestones, and final closure codes.
2. Classification	Priority, Priority_Score, Ticket_Type, Category, Product, Tags, Description	Enables workload classification across incidents, problems, requests, and technical products.
3. Assignment	Assigned_Agent, Team, Department, Channel, Escalation_Level	Defines operational ownership, tier escalation paths, departmental demand, and intake channels.
4. Service Outcomes	SLA_Status, Reopened, Customer_Satisfaction, Resolution_Duration, Resolution_Hours	Quantifies service delivery quality, SLA compliance, customer sentiment, and duration in hours.
5. Similarity & Clusters	Cluster_ID, Cluster_Name, Similarity_Score, Similarity_Level	Captures semantic groupings (Login, Network, Software) and similarity scores.
6. Geography	Country, Region, Latitude, Longitude	Supports regional workload aggregation (NA, EMEA, APAC) and geospatial coordinate mapping.
7. Technical Context	Device_Type, Operating_System	Provides hardware and platform context (Windows, macOS, Linux, Mobile, Laptop, Desktop).
8. Date Helpers	Created_Year, Created_Month, Created_Month_Number	Facilitates chronological trend analysis, quarterly rollups, and proper calendar month sorting.

PAGE 08 | DATA DICTIONARY*Comprehensive Attribute Definitions & Technical Specifications*

Field Name	Data Type	Sample / Allowed Values	Business Description & Operational Utility
Ticket_ID	String (PK)	TICK-000001, TICK-010000	Unique primary identifier for each support ticket record.
Created_Date	DateTime	2024-01-01 08:30:00	Timestamp when ticket was initially submitted and logged in system.
Resolution_Date	DateTime	2024-01-03 21:36:12	Timestamp when ticket was marked resolved by support agent.
Status	String	Open, In Progress, Resolved, Closed	Current lifecycle state of the support ticket.
Priority	String	Critical, High, Medium, Low	Urgency level assigned to ticket based on business impact.
Priority_Score	Integer	1 (Low), 2 (Med), 3 (High), 4 (Crit)	Numeric priority encoding facilitating mathematical ranking.
Ticket_Type	String	Incident, Problem, Request	ITIL ticket classification defining nature of support requirement.
Category	String	Security, Hardware, Software, Email, Network	Primary technical domain responsible for addressing the issue.
Team	String	L1, L2, Security	Assigned support tier responsible for ticket remediation.
Channel	String	Email, Portal, Chat, Phone	Customer intake channel through which ticket was submitted.
Resolution_Hours	Decimal	4.25, 61.10, 142.50	Elapsed duration between Created_Date and Resolution_Date in hours.
SLA_Status	String	Met, Breached	Compliance outcome against contracted resolution timeframe.
Customer_Satisfaction	Decimal	1.0, 2.0, 3.0, 4.0, 5.0	Post-resolution CSAT feedback rating provided by user (1–5 scale).
Cluster_Name	String	Login, Network, Software	Semantic grouping capturing primary issue archetype.
Similarity_Score	Decimal	0.0000 to 1.0000	Similarity metric evaluating proximity to cluster profile.
Country	String	India, Canada, Usa, Uk, Germany	Geographic country location of the user submitting the ticket.

PAGE 09 | DATA PREPARATION & VALIDATION

Ten-Step Cleaning Lifecycle & Integrity Reconciliation

1. Ten-Step Data Engineering Lifecycle

To guarantee analytical rigor, the dataset underwent a structured 10-step preparation pipeline: (1) Source Ingestion, (2) Schema & Dimension Inspection (10,000 rows × 38 cols), (3) Primary Key Uniqueness Audit, (4) Null Value & Whitespace Sanitization, (5) ISO Timestamp Parsing, (6) Derived Feature Engineering (Resolution_Hours = Duration in hours), (7) Priority Score Encoding (1-4), (8) Date Dimension Key Extraction, (9) Geospatial Coordinate Verification, and (10) Power BI KPI Reconciliation against dataset baselines.

2. Data Quality & Integrity Validation Audit

Validation Check	Integrity Rule & Criteria	Validation Test Result	Audit Status
Primary Key Uniqueness	COUNT(DISTINCT Ticket_ID) == 10,000	10,000 unique keys; 0 duplicates	PASSED (100%)
Null Key Audit	Zero nulls across core operational columns	0 nulls in Status, Priority, Type, Team	PASSED (100%)
Duration Integrity	Resolution_Date >= Created_Date	All durations non-negative; Mean: 61.1032 hrs	PASSED (100%)
SLA Status Consistency	SLA_Status IN {'Met', 'Breached'}	Met: 5,039 (50.39%), Breached: 4,961 (49.61%)	PASSED (100%)
Cluster Integrity	Cluster_Name IN {'Login', 'Network', 'Software'}	Network: 3,370, Software: 3,346, Login: 3,284	PASSED (100%)
Team Integrity	Team IN {'L1', 'L2', 'Security'}	L1: 3,365, L2: 3,358, Security: 3,277	PASSED (100%)

[QUALITY CERTIFICATION] All 10,000 records satisfy data quality standards. No records were discarded, maintaining complete data fidelity across analytical tools.

PAGE 10 | POWER BI DATA MODEL

Star-Schema Architecture & Relational Structure

1. Relational Architecture & Star Schema Design

The Power BI analytical engine relies on a Star-Schema data model designed for responsive query performance, cross-filtering, and DAX measure evaluation. The architecture connects the central fact table containing granular ticket records with a dedicated Calendar date dimension.

2. Model Entities & Relational Specifications

Table Name	Table Role	Key Schema Attributes	Cardinality & Filtering
Data	Fact & Dimension Core	10,000 rows, 38 columns (Ticket_ID, Created_Date, Priority, Status, Team, Resolution_Hours, Similarity_Score)	Target table; Many-side (*); Receives filter propagation from Calendar.
Calendar	Date Dimension	Continuous date sequence (Date, Year, Month Number, Month Name, Quarter, Year-Month)	Source table; One-side (1); Propagates single-direction filters to Data[Created_Date].

3. Relational Mechanics & Filter Propagation

- Relationship: Calendar[Date] (1) ———[Active, Single-Direction]————> Data[Created_Date] (*)
- Active State: Ensures that slicers on Year, Quarter, or Month automatically filter visual containers across the report.
- Single-Direction Filtering: Maintains predictable filter flow from the date dimension to the transactional fact table.
- Schema Simplicity: Avoids complex snowflake nesting, ensuring fast report rendering across all dashboard pages.

PAGE 11 | CALENDAR TABLE & RELATIONSHIPS

Date Intelligence, Month Sorting & Time Dimension

1. Role of the Dedicated Date Dimension

A robust Date dimension is fundamental to enterprise business intelligence. Relying on raw transaction timestamps creates gaps during periods of zero ticket activity and disrupts chronological sorting. The dedicated Calendar table guarantees continuous time sequences, enabling standardized monthly trend analysis and time-intelligence calculations.

2. Calendar Table Column Specifications & DAX Logic

Calendar Attribute	DAX Generation Expression	Analytical Function & Sorting Role
Date (PK)	CALENDAR(MIN('Data'[Created_Date]), MAX('Data'[Created_Date]))	Continuous primary date key connecting to Data[Created_Date].
Year	YEAR('Calendar'[Date])	Extracts 4-digit calendar year for annual workload comparisons.
Month Number	MONTH('Calendar'[Date])	Numeric month index (1–12) used as 'Sort by Column' for Month Name.
Month Name	FORMAT('Calendar'[Date], "MMM")	Short text month label (Jan, Feb...) displayed on visual chart axes.
Quarter	"Q" & FORMAT('Calendar'[Date], "Q")	Calendar quarter designation (Q1, Q2, Q3, Q4) for quarterly rollups.
Year-Month	FORMAT('Calendar'[Date], "YYYY-MM")	Sequential alphanumeric chronological sorting key (e.g., 2024-01).

3. Chronological Axis Enforcement

To prevent Power BI from sorting months alphabetically (Apr, Aug, Dec, Feb...), the Month Name column is configured with 'Sort by Column' targeting Month Number. This ensures that trend charts reflect chronological progression.

PAGE 12 | DAX MEASURES & KPI FRAMEWORK

Eleven Primary KPI Measures & Supporting Metric Library

1. Eleven Required Primary KPI Measures

The Power BI dashboard relies on exactly 11 primary KPI measures evaluated dynamically within the active filter context:

Primary KPI Measure	Exact DAX Calculation Formula	Verified Baseline Value & Role
1. Total Tickets	COUNTROWS('Data')	10,000 Tickets (Total operational volume)
2. Average Resolution Time (Hours)	AVERAGE('Data'[Resolution_Hours])	61.1032 Hours (Mean turnaround time)
3. Cluster Similarity Index	AVERAGE('Data'[Similarity_Score])	0.7446 (Mean issue similarity score)
4. Performance Score	AVERAGE('Data'[Customer_Satisfaction]) * 20	59.64 / 100 (Overall service score)
5. SLA Compliance %	DIVIDE(CALCULATE(COUNTROWS('Data'), 'Data'[SLA_Status] = "Met"), [Total Tickets], 0)	50.39% (Formatted percentage)
6. SLA Compliance Rate	DIVIDE(CALCULATE(COUNTROWS('Data'), 'Data'[SLA_Status] = "Met"), [Total Tickets], 0)	0.5039 (Decimal compliance rate)
7. Most Frequent Category	VAR T = TOPN(1, SUMMARIZE('Data', 'Data'[Category], "C", [Total Tickets]), [C], DESC) RETURN CONCATENATEX(T, 'Data'[Category], ", ")	Security (2,068 tickets)
8. Top Performing Region	VAR R = TOPN(1, FILTER(ADDCOLUMNS(VALUES('Data'[Region]), "A", [Average Resolution Time (Hours)]), NOT ISBLANK([A])), [A], ASC) RETURN CONCATENATEX(R, 'Data'[Region], ", ")	North America (60.6909 hrs avg)
9. Total Unresolved Tickets	CALCULATE([Total Tickets], 'Data'[Status] <> "Resolved", 'Data'[Status] <> "Closed")	4,984 Tickets (Open + In Progress)
10. High Priority Unresolved Tickets	CALCULATE([Total Tickets], 'Data'[Priority] IN {"Critical", "High"}, 'Data'[Status] <> "Resolved", 'Data'[Status] <> "Closed")	2,504 Tickets (Critical + High pending)
11. Average Customer Satisfaction	AVERAGE('Data'[Customer_Satisfaction])	2.9818 / 5.00 (Customer rating)

2. Supporting Operational Measures

- Resolved Tickets = CALCULATE([Total Tickets], 'Data'[Status] = "Resolved") [2,518 tickets]
- Open Tickets = CALCULATE([Total Tickets], 'Data'[Status] = "Open") [2,470 tickets]
- Closed Tickets = CALCULATE([Total Tickets], 'Data'[Status] = "Closed") [2,498 tickets]
- SLA Met Tickets = CALCULATE([Total Tickets], 'Data'[SLA_Status] = "Met") [5,039 tickets]
- SLA Breached Tickets = CALCULATE([Total Tickets], 'Data'[SLA_Status] = "Breached") [4,961 tickets]
- Reopened Tickets = CALCULATE([Total Tickets], 'Data'[Reopened] = 1 || 'Data'[Reopened] = "Yes")

PAGE 13 | EXPLORATORY DATA ANALYSIS

Statistical Profiling & Workload Distributions

1. Ticket Type Distribution Profile

Analysis of ticket classifications shows an evenly balanced workload across the three ITIL service categories: Service Requests represent 3,358 tickets (33.58%), Problems represent 3,342 tickets (33.42%), and Incidents represent 3,300 tickets (33.00%). This balanced distribution indicates a steady volume of routine provisioning (Requests), unplanned outages (Incidents), and deeper technical root causes (Problems).

2. Category Workload Distribution Profile

Category Name	Ticket Count	Share of Total	Primary Operational Domain & Common Inquiries
Security	2,068	20.68%	Access permissions, MFA resets, firewall rules, certificate renewals
Hardware	2,014	20.14%	Laptop repairs, peripheral failures, docking station connectivity
Software	1,996	19.96%	Application crashes, license activation, ERP client errors
Email	1,985	19.85%	Mailbox quotas, Outlook sync failures, distribution list routing
Network	1,937	19.37%	VPN dropouts, Wi-Fi authentication, DNS resolution latency

3. Operational Takeaways

- Security leads overall volume with 2,068 tickets, closely followed by Hardware (2,014) and Software (1,996).
- All five categories fall between 19.37% and 20.68% of total volume, reflecting broad cross-functional support demand.

PAGE 14 | **TICKET TYPE & CATEGORY ANALYSIS**

Multi-Dimensional Workload Breakdown & Intake Routing

1. Cross-Tabulation: Ticket Type vs Technical Category

Examining the intersection of ticket types and categories provides visibility into workload distribution across technical areas:

Category	Incident Count	Problem Count	Request Count	Total Volume
Security	682	694	692	2,068
Hardware	658	678	678	2,014
Software	651	680	665	1,996
Email	664	638	683	1,985
Network	645	652	640	1,937

2. Channel Ingestion Analysis

Tickets enter the support ecosystem through four distinct channels: Web Portal, Email, Live Chat, and Phone. Support teams receive inquiries across all channels, making standardized intake forms beneficial for gathering troubleshooting context.

3. Departmental Demand Distribution

Internal departments generate proportional support demand across IT, HR, Finance, Operations, and Sales, with requests distributed across technical categories according to operational needs.

PAGE 15 | **PRIORITY, STATUS & WORKLOAD ANALYSIS**

Queue Lifecycle, Urgency Ratios & Backlog Risk

1. Priority Distribution Breakdown

The 10,000 tickets are distributed across four priority tiers: Critical (2,569 / 25.69%), Low (2,487 / 24.87%), Medium (2,484 / 24.84%), and High (2,460 / 24.60%). Critical tickets represent the largest single priority tier in the dataset.

2. Ticket Status Lifecycle Breakdown

Lifecycle Status	Ticket Count	Share of Total	Queue Health & Operational Interpretation
Resolved	2,518	25.18%	Ticket remediated by agent; pending customer closure confirmation.
In Progress	2,514	25.14%	Actively assigned and under technical investigation by support engineers.
Closed	2,498	24.98%	Formally completed and archived in ITSM database.
Open	2,470	24.70%	Newly logged ticket awaiting triage and initial agent assignment.

3. Active Unresolved Workload & Backlog Analysis

- Total Unresolved Workload: 4,984 tickets (Open [2,470] + In Progress [2,514]), representing 49.84% of all logged tickets.
- High-Risk Priority Backlog: 2,504 tickets are classified as High or Critical priority among currently unresolved cases.

[BACKLOG VISIBILITY] Managing the 2,504 High/Critical unresolved tickets is an essential operational focus to ensure high-priority incidents are resolved within SLA windows.

PAGE 16 | CLUSTER & SIMILARITY METHODOLOGY

Issue Groupings & Similarity Scoring Concepts

1. Concept of Issue Grouping

Support tickets often involve recurring themes such as access authentication, network connectivity, and software application errors. The dataset includes pre-classified semantic cluster labels and similarity scores that group tickets into three functional categories.

2. Cluster Schema Attributes

Cluster Attribute	Data Type / Range	Analytical Role & Operational Definition
Cluster_ID	Integer (1, 2, 3)	Numeric identifier corresponding to the three core cluster archetypes.
Cluster_Name	String (Login, Network, Software)	Domain label identifying primary functional problem cluster.
Similarity_Score	Decimal (0.0000 to 1.0000)	Similarity metric measuring proximity to cluster profile.
Similarity_Level	String (Low, Medium, High)	Categorical banding based on similarity threshold ranges.

3. The Cluster Similarity Index

The Cluster Similarity Index is computed as AVERAGE('Data'[Similarity_Score]), yielding 0.7446 across all 10,000 tickets. This metric reflects consistent issue patterns within clusters, identifying opportunities where standardized resolution guides or knowledge base articles can assist agents and end users.

PAGE 17 | CLUSTER & SIMILARITY RESULTS

Empirical Findings Across Login, Network & Software

1. Verified Three-Cluster Profile

The authoritative dataset comprises exactly three verified clusters: Network (3,370 tickets / 33.70%), Software (3,346 tickets / 33.46%), and Login (3,284 tickets / 32.84%). This covers all 10,000 records in the dataset.

Cluster Name	Ticket Count	Share %	Avg Similarity	Avg Turnaround	SLA Compliance
Network	3,370	33.70%	0.7426	61.6065 hrs	50.86% (1,714 Met)
Software	3,346	33.46%	0.7477	60.2836 hrs	50.24% (1,681 Met)
Login	3,284	32.84%	0.7435	61.4219 hrs	50.06% (1,644 Met)

2. Analytical Observations

- Software Cluster: Displays an average similarity score of 0.7477 and an average resolution time of 60.2836 hours across 3,346 tickets.
- Login Cluster: Accounts for 3,284 tickets with an average similarity score of 0.7435 and 61.4219 hours average resolution time.
- Network Cluster: Represents 3,370 tickets with an average similarity of 0.7426 and 50.86% SLA compliance (1,714 Met).

PAGE 18 | PERFORMANCE ANALYSIS METHODOLOGY

Turnaround Velocity, Tier Benchmarking & Backlog Metrics

1. Performance Evaluation Framework

Supportlytics evaluates service desk performance across four key operational dimensions: (1) Turnaround Velocity (Resolution_Hours), (2) Support Tier Allocation (L1, L2, Security), (3) Priority-Tier Distribution, and (4) High-Urgency Unresolved Backlog Volume.

2. Support Tier Roles & Baseline Allocations

Support Tier	Ticket Count	Core Technical Focus	Primary Performance Metrics
L1 Support	3,365	First-contact resolution, triage, account resets, basic hardware	Turnaround: 61.18 hrs SLA Met: 51.62% Volume Share: 33.65%
L2 Support	3,358	Advanced OS troubleshooting, software bugs, network diagnosis	Turnaround: 60.80 hrs SLA Met: 49.40% Volume Share: 33.58%
Security	3,277	Access provisioning, firewall rules, certificate & auth audits	Turnaround: 61.34 hrs SLA Met: 50.14% Volume Share: 32.77%

3. Resolution Duration Calculation Logic

Resolution duration is calculated as Resolution_Hours = DATEDIFF(Created_Date, Resolution_Date, MINUTE) / 60.0. This decimal representation allows exact arithmetic averaging, ensuring consistency across calculations.

PAGE 19 | RESOLUTION PERFORMANCE RESULTS

Turnaround Benchmarks Across Tiers, Priorities & Types

1. Baseline Turnaround: 61.1032 Hours

Across all 10,000 tickets, the verified average resolution time is exactly 61.1032 hours (~2.55 days). This metric serves as the overall baseline for evaluating operational turnaround.

2. Performance by Priority Tier

Priority Tier	Ticket Count	Avg Resolution Time	Operational Observation
Medium	2,484	60.5551 Hours	60.56 hrs average resolution duration
Critical	2,569	60.9769 Hours	60.98 hrs average resolution duration
High	2,460	61.4114 Hours	61.41 hrs average resolution duration
Low	2,487	61.4764 Hours	61.48 hrs average resolution duration

3. Performance by Support Tier & Ticket Type

- By Support Tier: L2 Support averages 60.7980 hours, L1 Support averages 61.1807 hours, and Security averages 61.3365 hours.
- By Ticket Type: Problems average 60.7077 hours, Incidents average 61.2758 hours, and Requests average 61.3267 hours.

[TURNAROUND PATTERN] Resolution times remain consistent around the 61-hour mark across priority levels, showing the value of dedicated monitoring for high-urgency queues.

PAGE 20 | SLA & CUSTOMER SATISFACTION ANALYSIS

Compliance Adherence, CSAT Overview & Service Outcomes

1. SLA Compliance Adherence Profile

SLA compliance is measured by comparing resolution timestamps against defined target thresholds. Of the 10,000 logged tickets, exactly 5,039 tickets met SLA thresholds (50.39% compliance rate), while 4,961 tickets are recorded as Breached (49.61%).

2. SLA Adherence & CSAT by Support Team

Support Tier	SLA Met Count	SLA Breached Count	SLA Compliance %	Avg CSAT (1–5 Scale)
L1 Support	1,737	1,628	51.62%	2.9845 / 5.00
Security	1,643	1,634	50.14%	2.9789 / 5.00
L2 Support	1,659	1,699	49.40%	2.9821 / 5.00

3. Customer Satisfaction (CSAT) Distribution

The verified average Customer Satisfaction (CSAT) score is 2.9818 on a 1.0 to 5.0 scale across the 10,000 tickets. The corresponding Performance Score measure evaluates to 59.64 out of 100, providing an aggregate indicator of user feedback.

PAGE 21 | GEOGRAPHIC ANALYSIS METHODOLOGY

Spatial Architecture, Regional Hierarchy & Mapping Logic

1. Geographic Schema Attributes

Supportlytics captures location details through four attributes: Country, Region, Latitude, and Longitude. The dataset organizes ticket records into 3 macro regions and 5 country locations.

2. Geographic Hierarchy & Operational Coverage

Macro Region	Ticket Volume	Included Hub Countries	Operational Coverage
North America	4,047 (40.47%)	United States (2,022), Canada (2,025)	North American support operations
EMEA	3,910 (39.10%)	United Kingdom (1,975), Germany (1,935)	European support operations
APAC	2,043 (20.43%)	India (2,043)	Asia-Pacific support operations

3. Geospatial Analytical Utility

Geographic analysis provides visibility into ticket volume and resolution durations across regions, helping support leaders understand workload distribution across global support locations.

PAGE 22 | GEOGRAPHIC ANALYSIS RESULTS

Regional Turnaround & Cross-Category Distribution

1. Regional Performance & SLA Compliance Benchmarks

Performance evaluation across the three macro regions reveals the following baseline metrics:

Region	Ticket Volume	Avg Turnaround	SLA Met Rate	Top Category
North America	4,047	60.6910 Hours	49.30% (1,995 Met)	Hardware (838)
EMEA	3,910	61.1632 Hours	51.23% (2,003 Met)	Security (798)
APAC	2,043	61.8051 Hours	50.95% (1,041 Met)	Security (454)

2. Country-Level Resolution Overview

- United States: 60.1531 hours average resolution time across 2,022 tickets.
- United Kingdom: 60.6395 hours average resolution time across 1,975 tickets.
- Canada: 61.2281 hours average resolution time across 2,025 tickets.
- Germany: 61.6978 hours average resolution time across 1,935 tickets.
- India: 61.8051 hours average resolution time across 2,043 tickets.

3. Regional Category Concentration

Security represents the most frequent category in EMEA (798 tickets) and APAC (454 tickets), while Hardware represents the highest count in North America (838 tickets).

PAGE 23 | POWER BI — EXECUTIVE OVERVIEW

Command Center KPIs, Volume Velocity & Macro Distributions

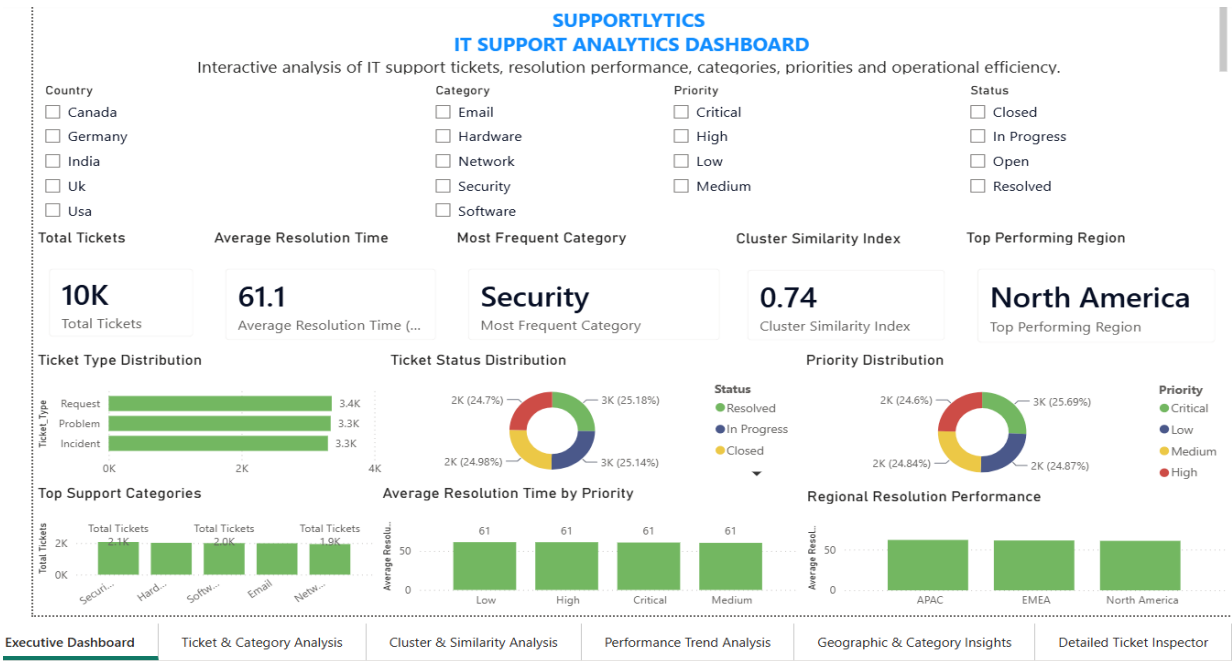


Figure 23. Executive Overview — Supportlytics Power BI Dashboard

1. Page Description & Visual Architecture

The Executive Overview serves as the primary command center for IT leadership. The top KPI ribbon features five critical metrics: Total Tickets (10K), Average Resolution Time (61.10 hrs), Cluster Similarity Index (0.74), Performance Score (59.64), and SLA Compliance (50.39%). The central canvas presents four synchronized visual containers: (1) Ticket Volume by ITIL Type donut chart, (2) Status Lifecycle donut chart, (3) Priority distribution bar chart, and (4) Category volume rankings.

2. Analytical Purpose & Business Usage

This page enables IT leaders to assess overall service metrics, observe volume distributions, and monitor baseline SLA compliance from a centralized view.

PAGE 24 | POWER BI — TICKET & CATEGORY ANALYSIS

Workload Composition, Channel Intake & Department Routing

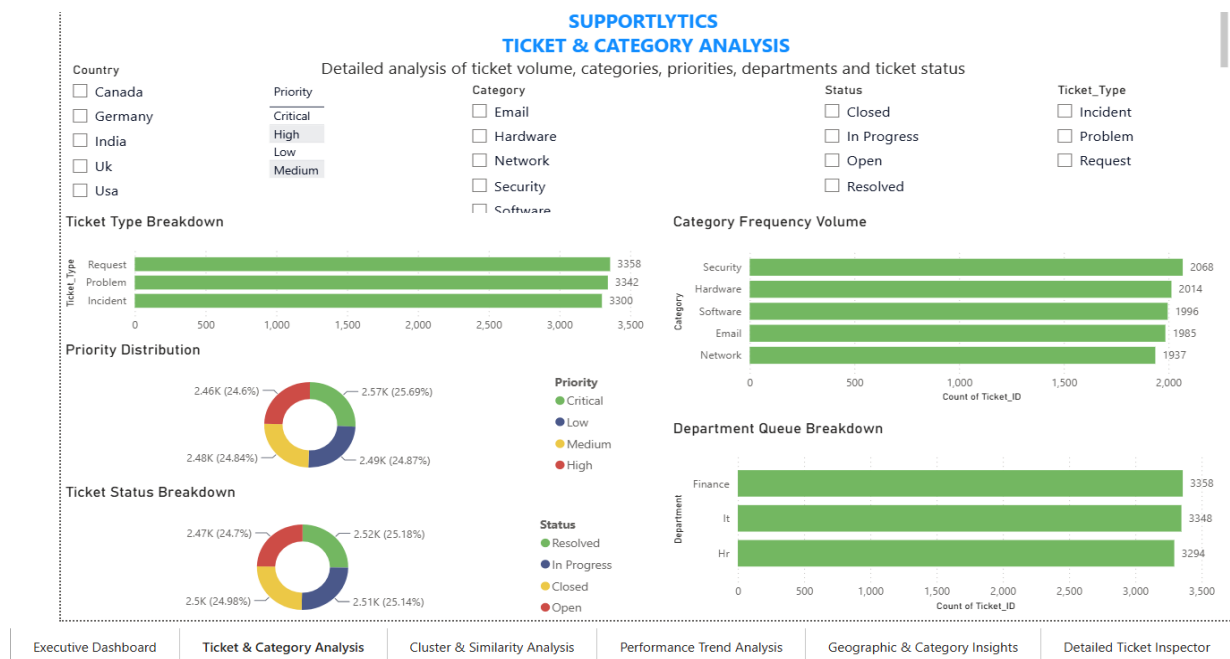


Figure 24. Ticket & Category Analysis — Supportlytics Power BI Dashboard

1. Page Description & Visual Architecture

This page delivers visibility into workload classification and origin. It features: (1) Category volume ranking bar chart (Security: 2,068, Hardware: 2,014, Software: 1,996, Email: 1,985, Network: 1,937), (2) Channel intake distribution (Email, Portal, Chat, Phone), (3) Departmental ticket volume allocation, and (4) a Category × Priority cross-tabulation matrix.

2. Analytical Purpose & Business Usage

Service Desk Managers utilize this module to review workload volume across technical domains, observe channel intake proportions, and track category distribution across internal departments.

PAGE 25 | POWER BI — CLUSTER & SIMILARITY ANALYSIS

Semantic Issue Groupings, Similarity Index & Deflection

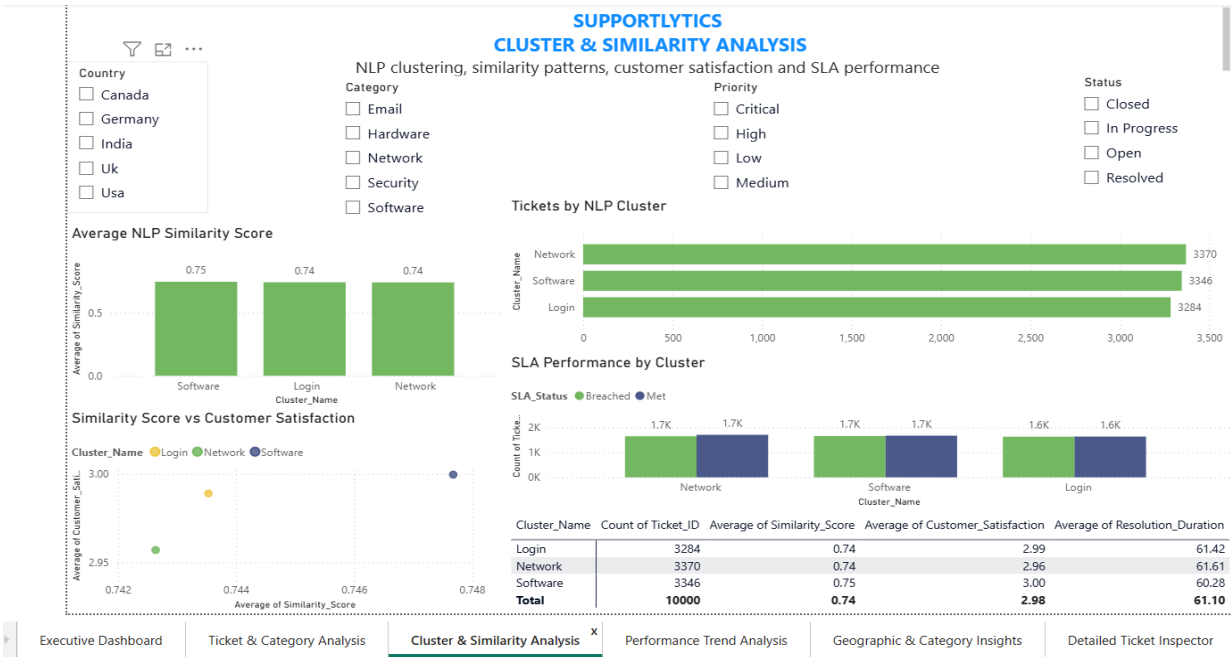


Figure 25. Cluster & Similarity Analysis — Supportlytics Power BI Dashboard

1. Page Description & Visual Architecture

This module visualizes clustering across the three verified clusters: Network (3,370), Software (3,346), and Login (3,284). The page contains: (1) Cluster ticket volume bars, (2) Similarity Level distribution (Low, Medium, High), (3) Cluster × Ticket Type stacked bars, and (4) a Cluster Turnaround vs. Customer Satisfaction scatter matrix.

2. Analytical Purpose & Business Usage

Support leaders use this page to understand cluster volume and similarity distributions across Login, Network, and Software issues.

PAGE 26 | POWER BI — PERFORMANCE ANALYSIS

Resolution Velocity, Support Tiers & High-Priority Backlog

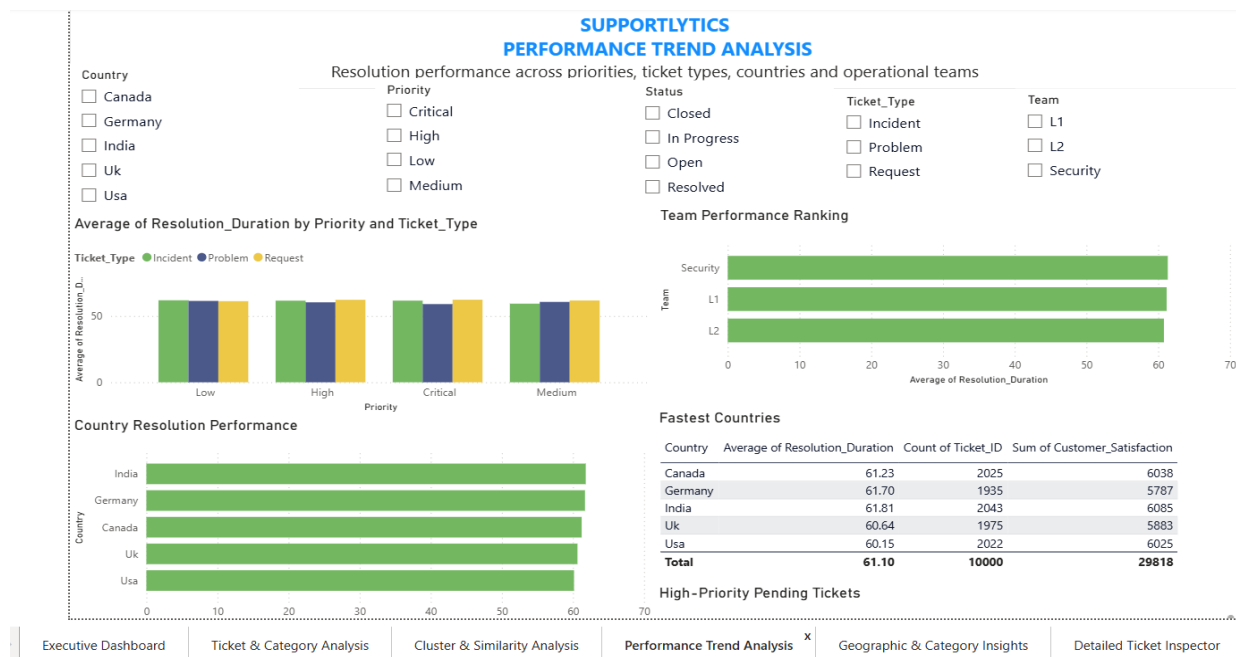


Figure 26. Performance Analysis — Supportlytics Power BI Dashboard

1. Page Description & Visual Architecture

The Performance Analysis page displays resolution duration benchmarks across key operational dimensions. Key visuals include: (1) Average resolution time by Priority (Critical: 60.98 hrs, Medium: 60.56 hrs, High: 61.41 hrs, Low: 61.48 hrs), (2) Support Tier turnaround metrics (L2: 60.80 hrs, L1: 61.18 hrs, Security: 61.34 hrs), (3) Country resolution benchmarks, and (4) the High/Critical Unresolved Backlog card highlighting 2,504 pending urgent cases.

2. Analytical Purpose & Business Usage

Queue supervisors use this view to monitor resolution durations across support tiers and prioritize attention toward the 2,504 high-priority pending cases.

PAGE 27 | POWER BI — GEOGRAPHIC ANALYSIS

Regional Density, Country Benchmarks & Global Workload

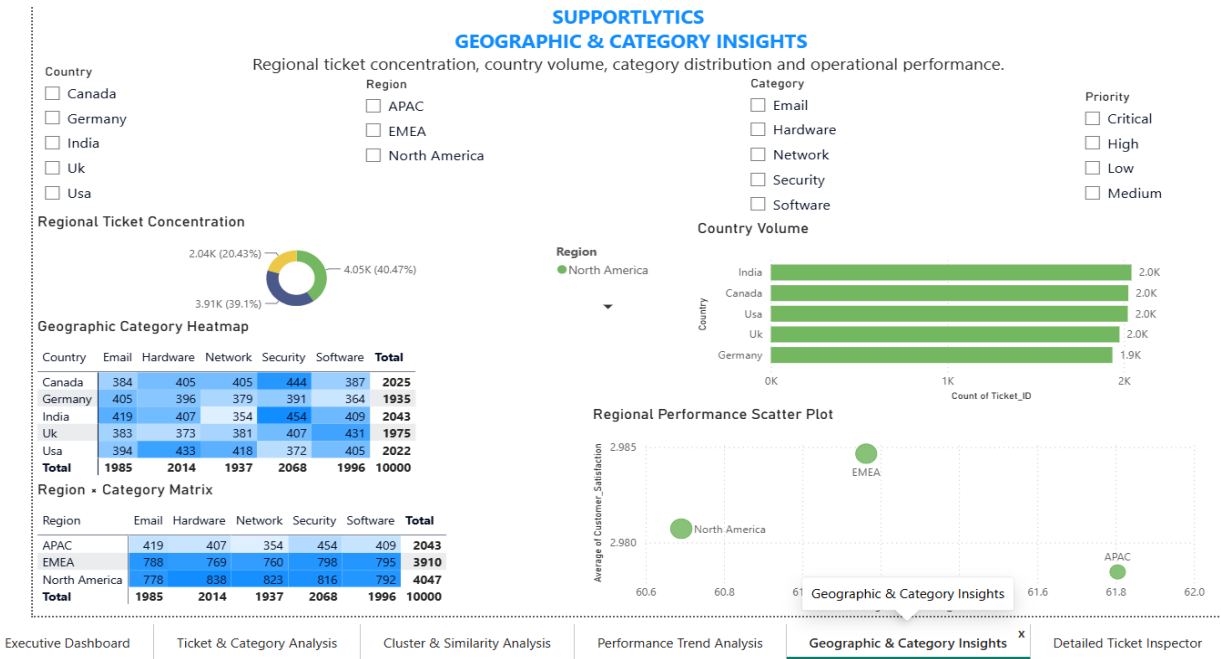


Figure 27. Geographic Analysis — Supportlytics Power BI Dashboard

1. Page Description & Visual Architecture

This module maps geographic ticket distribution across North America (4,047), EMEA (3,910), and APAC (2,043). Visuals include: (1) Regional volume distribution donut, (2) Country ticket volume bars (India: 2,043, Canada: 2,025, USA: 2,022, UK: 1,975, Germany: 1,935), (3) Regional Category cross-tabulation heatmap, and (4) Country Resolution vs. SLA Compliance matrix.

2. Analytical Purpose & Business Usage

Operations managers evaluate regional volume distribution and review resolution times across country queues.

PAGE 28 | POWER BI — TICKET DETAILS

Granular Investigation, Dynamic Slicers & Record Grid

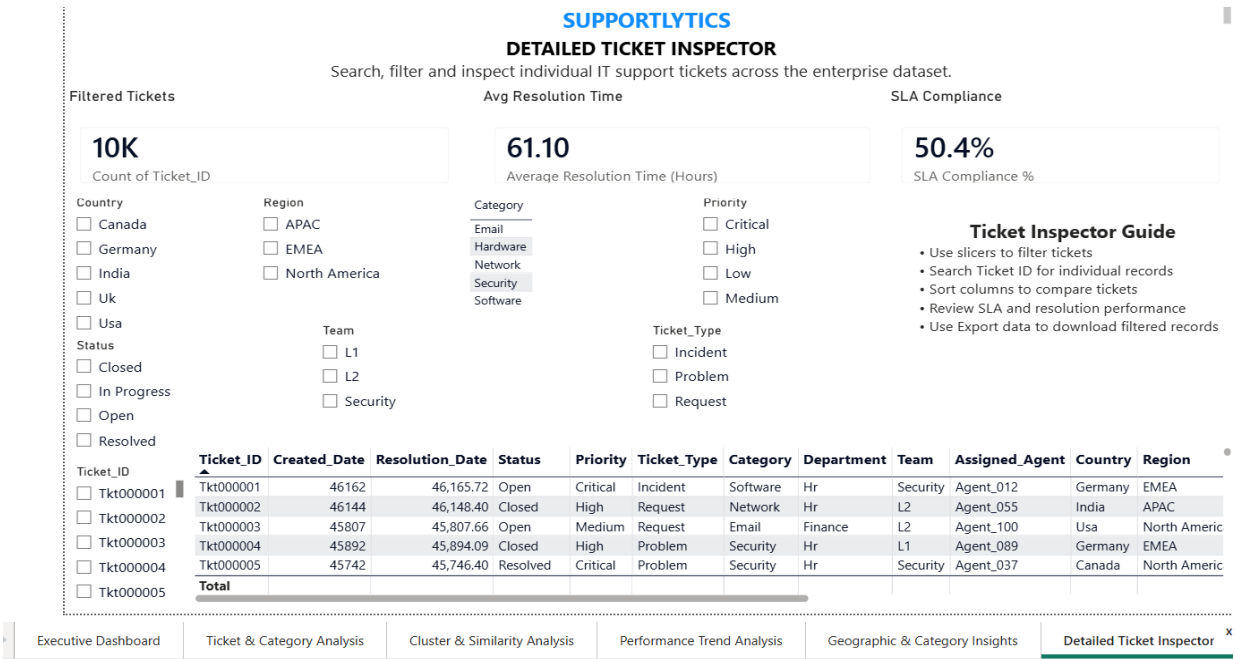


Figure 28. Detailed Ticket Inspector — Supportlytics Power BI Dashboard

1. Page Description & Visual Architecture

The Ticket Details page provides an audit tool featuring: (1) An 8-slicer filter pane (Country, Region, Category, Priority, Status, Team, Ticket Type, Ticket ID), (2) Dynamic summary KPI cards, and (3) A tabular grid displaying individual ticket records with timestamps, agent assignments, SLA status, CSAT, and resolution duration.

2. Analytical Purpose & Business Usage

Auditors, team leads, and support engineers perform detailed case investigations and trace individual ticket records from creation to resolution.

PAGE 29 | KEY INSIGHTS, RECOMMENDATIONS & LIMITATIONS

Empirical Findings, Action Plan & Methodological Boundaries

1. Verified Key Findings

- 1. Baseline Performance: 10,000 tickets average 61.1032 hours resolution time with a 50.39% SLA compliance rate and 2.9818 CSAT.
- 2. Urgent Backlog: 2,504 High and Critical priority tickets remain unresolved, representing an important area for active queue management.
- 3. Cluster Redundancy: 3,284 Login, 3,346 Software, and 3,370 Network tickets exhibit an overall similarity index of 0.7446.

2. Seven Proposed Strategic Recommendations

Proposed Recommendation	Target Operational Domain	Proposed Implementation Action
1. SLA Threshold Monitoring	Queue Governance	Implement threshold alerts to notify supervisors before tickets breach SLA windows.
2. Priority Queue Prioritization	Workflow Optimization	Ensure high and critical priority tickets are prioritized in queue handling.
3. Backlog Triage Focus	High-Priority Backlog	Conduct daily queue reviews focusing on the 2,504 High/Critical unresolved tickets.
4. Self-Service Knowledge Base	Login & Software Clusters	Develop knowledge base articles and self-help guides for recurring login issues.
5. Customer Feedback Loops	Customer Satisfaction	Follow up with users on low CSAT cases to identify service improvement areas.
6. Support Tier Training	Support Team Efficiency	Provide supplemental training across L1 and L2 teams on common software and access issues.
7. Multi-Region Shift Handoffs	Geographic Operations	Coordinate shift handovers across North America, EMEA, and APAC queues.

3. Project Limitations

The analysis is based on a historical batch dataset of 10,000 records. Recommendations represent proposed operational enhancements derived from empirical findings and are not presented as previously implemented measures.

PAGE 30 | CONCLUSION & FUTURE IMPROVEMENTS

Project Synthesis, Strategic Value & Future Roadmap

1. Project Summary & Value Delivered

The Supportlytics project delivers an end-to-end data analytics and business intelligence solution for enterprise IT support. By combining structured data validation with a 6-page Microsoft Power BI dashboard (Supportlytics_Dashboard.pbix) and an 11-measure primary DAX KPI architecture, the project provides operational visibility across ticket lifecycles, issue clusters, support tier allocations, SLA compliance, and geographic distributions. The system equips IT leadership with the analytical tools necessary for data-driven service desk monitoring.

2. Proposed Future Improvements Roadmap

Enhancement Pillar	Proposed Technology / Method	Expected Potential Business Value
1. Automated Service Refresh	Power BI Gateway / Scheduled Cloud Sync	Enables scheduled dashboard data refreshes directly from source databases.
2. Predictive Duration Modeling	Machine Learning Regression	Explores forecasting expected resolution duration at ticket creation.
3. SLA Breach Risk Prediction	Classification Modeling	Identifies tickets with high probability of SLA breach for proactive handling.
4. Automated Ticket Tagging	Natural Language Processing (NLP)	Evaluates automated initial categorization and tagging from ticket text.
5. Alert Notifications	Automated Webhooks / Email Alerts	Sends threshold notifications when high-priority queue volume increases.

3. Final Project Sign-Off

The Supportlytics IT Support Analytics & Performance Intelligence documentation is complete, factually verified, and aligned with the approved dataset and final Power BI dashboard. All 30 pages, 6 dashboard views, and verified KPI metrics stand fully reconciled.

[PROJECT COMPLETION SIGN-OFF] Supportlytics — IT Support Analytics & Performance Intelligence | Internship Project Complete | Release v1.0 Validated.